

SWT 22022-Practical for  
Internet Application  
Development

Department of Information  
and Communication  
Technology  
Faculty of Technology



**Lab sheet :09**  
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**Practical No :09**

# Title: PHP, MySQL and JavaScript - II

## Aims:

- Implement CRUD operations with PHP, MySQL, and JavaScript

## Tasks:

- Implement CRUD operations in PHP
- Use fetch() to handle frontend requests

## Activities :

1. Create a new table named “**expenses**” within “**expense\_tracker**” database
2. Implement CRUD operation in PHP  
I.add\_expense.php

```
<?php
    require_once '../database/Expense.php';
    require_once '../database/Database.php';
    $database = new Database("localhost", "root", "", "expense_tracker");
    $db = $database->getConnection();
    $expense = new Expense($db);
    $data = json_decode(file_get_contents("php://input"));
    $expense->setDescription($data->description);
    $expense->setAmount($data->amount);
    if ($expense->create()) {
        echo json_encode(["message" => "Expense added."]);
    } else {
        echo json_encode(["message" => "Unable to add expense."]);
    }
?>
```

II.view\_expenses.php

```
<?php
    require_once '../database/Expense.php';
    require_once '../database/Database.php';
    $database = new Database("localhost", "root", "", "expense_tracker");
    $db = $database->getConnection();
    $expense = new Expense($db);
```

```

    $expenses = $expense->getAllExpenses();
    $num = count($expenses);
    if ($num > 0) {
        $expenses_arr = array();
        $expenses_arr["records"] = $expenses;
        echo json_encode($expenses_arr);
    } else {
        echo json_encode(["records" => []]);
    }
}
?>

```

### III.edit\_expense.php

```

<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';
$dbase = new Database("localhost", "root", "", "expense_tracker");
$db = $dbase->getConnection();
$expense = new Expense($db);
$data = json_decode(file_get_contents("php://input"));
$expense->setId($data->id);
$expense->setDescription($data->description);
$expense->setAmount($data->amount);
if ($expense->update()) {
    echo json_encode(["message" => "Expense updated."]);
} else {
    echo json_encode(["message" => "Unable to update expense."]);
}
?>

```

### IV.delete\_expense.php

```

<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';
$dbase = new Database("localhost", "root", "", "expense_tracker");
$db = $dbase->getConnection();
$expense = new Expense($db);
$data = json_decode(file_get_contents("php://input"));
$expense->setId($data->id);
if ($expense->delete()) {
    echo json_encode(["message" => "Expense deleted."]);
} else {
    echo json_encode(["message" => "Unable to delete expense."]);
}
?>

```

### 3. Use fetch() to handle frontend requests

#### I. loadExpenses()

```
async function loadExpenses() {
  try {
    const response = await fetch('php/view_expenses.php');
    const data = await response.json();
    const expenseList = document.getElementById('expenseList');
    expenseList.innerHTML = '';
    if (Array.isArray(data.records) && data.records.length > 0) {
      data.records.forEach(expense => {const item =
document.createElement('li');
      item.innerHTML = `${expense.created_at}:
      ${expense.description} - ${expense.amount}
      <button
onclick="editExpense(${expense.id}, '${expense.description}',
${expense.amount}, '${expense.created_at}')">Edit</button>
      <button onclick="deleteExpense(${expense.id})">Delete</button>`;
      expenseList.appendChild(item);
    });
  } else {const item = document.createElement('li');
    item.innerHTML = 'No expenses found.';
    expenseList.appendChild(item);
  }
} catch (error) {
  console.error('Error loading expenses:', error);
}
}
```

#### II. addExpense()

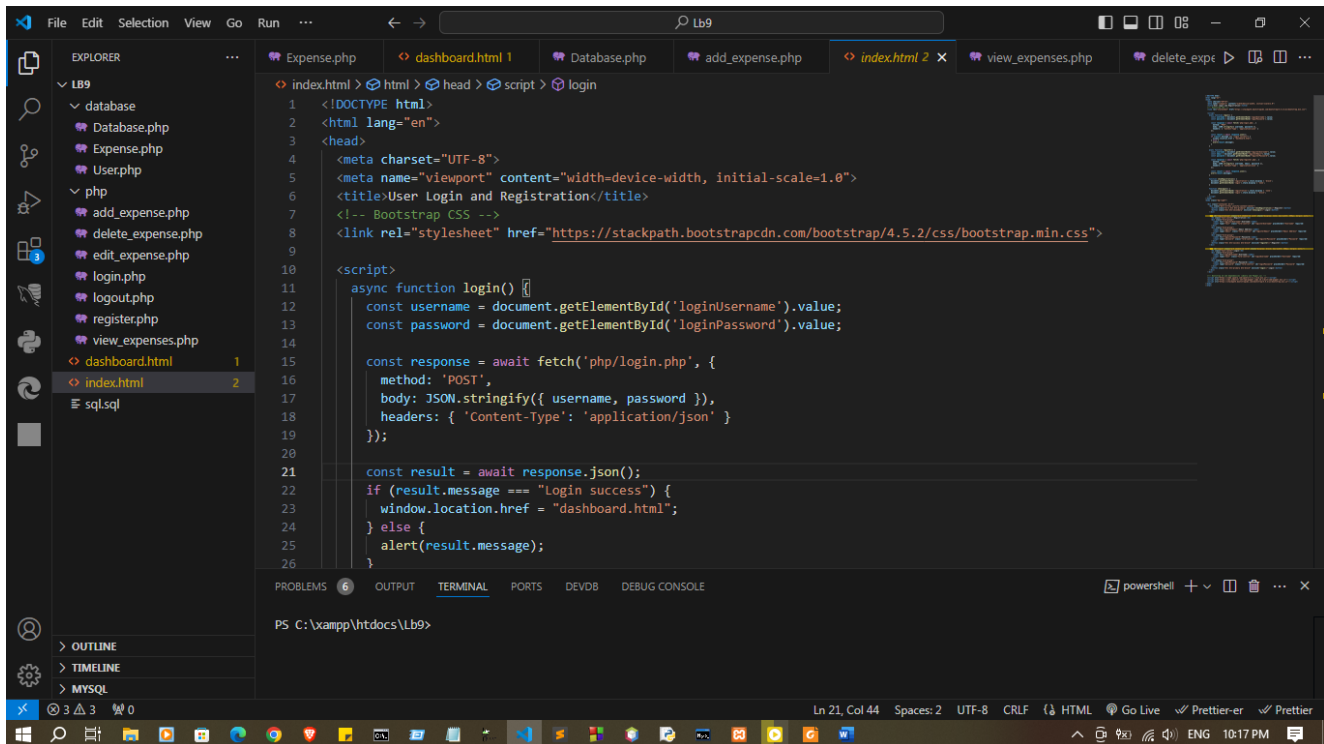
```
async function addExpense() {
  const description =
document.getElementById('description').value;
  const amount = document.getElementById('amount').value;
  const response = await fetch('php/add_expense.php', {
    method: 'POST',
    body: JSON.stringify({ description, amount }),
    headers: { 'Content-Type': 'application/json' }
  });
  const result = await response.json();
  alert(result.message);
  loadExpenses();
}
```

### III. updateExpense()

```
async function updateExpense() {
  const id = document.getElementById('expenseId').value;
  const description = document.getElementById('editDescription').value;
  const amount = document.getElementById('editAmount').value;
  const response = await fetch('php/edit_expense.php', {
    method: 'POST',
    body: JSON.stringify({ id, description, amount }),
    headers: { 'Content-Type': 'application/json' }
  });
  const result = await response.json();
  alert(result.message);
  document.getElementById('editForm').style.display = 'none';
  loadExpenses();
}
```

### IV. deleteExpense()

```
async function updateExpense() {
  const id = document.getElementById('expenseId').value;
  const description =
  document.getElementById('editDescription').value;
  const amount = document.getElementById('editAmount').value;
  const response = await fetch('php/edit_expense.php', {
    method: 'POST',
    body: JSON.stringify({ id, description, amount }),
    headers: { 'Content-Type': 'application/json' }
  });
  const result = await response.json();
  alert(result.message);
  document.getElementById('editForm').style.display = 'none';
  loadExpenses();
}
```



index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Login and Registration</title>
  <!-- Bootstrap CSS -->
  <link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">

  <script>
    async function login() {
      const username = document.getElementById('loginUsername').value;
      const password = document.getElementById('loginPassword').value;

      const response = await fetch('php/login.php', {
        method: 'POST',
        body: JSON.stringify({ username, password }),
        headers: { 'Content-Type': 'application/json' }
      });

      const result = await response.json();
      if (result.message === "Login success") {
        window.location.href = "dashboard.html";
      } else {

```

```

        alert(result.message);
    }
}

async function register() {
    const username = document.getElementById("registerUsername").value;
    const email = document.getElementById("registerEmail").value;
    const password = document.getElementById("registerPassword").value;

    const response = await fetch('php/register.php', {
        method: 'POST',
        body: JSON.stringify({ username, email, password }),
        headers: { 'Content-Type': 'application/json' }
    });

    const result = await response.json();
    alert(result.message);
}

function showRegistration() {
    document.getElementById('registration').style.display = 'block';
    document.getElementById('login').style.display = 'none';
}

function showLogin() {
    document.getElementById('registration').style.display = 'none';
    document.getElementById('login').style.display = 'block';
}
</script>
</head>
<body class="bg-light">

    <div class="container mt-5">
        <div class="mb-3 d-flex justify-content-center">
            <button class="mr-2 btn btn-primary"
onclick="showRegistration()">Register</button>
            <button class="btn btn-secondary" onclick="showLogin()">Login</button>
        </div>

        <div id="registration" class="p-4 shadow-sm card" style="display: none; max-
width: 400px; margin: auto;">
            <h2 class="text-center">Registration</h2>
            <div class="form-group">
                <label for="registerUsername">Username</label>
                <input type="text" class="form-control" id="registerUsername"
placeholder="Username" required>
            </div>
            <div class="form-group">

```

```

        <label for="registerEmail">Email Address</label>
        <input type="email" class="form-control" id="registerEmail"
placeholder="Email Address" required>
    </div>
    <div class="form-group">
        <label for="registerPassword">Password</label>
        <input type="password" class="form-control" id="registerPassword"
placeholder="Password" required>
    </div>
    <button class="btn btn-success btn-block"
onclick="register()">Register</button>
</div>

    <div id="login" class="p-4 shadow-sm card" style="display: block; max-width:
400px; margin: auto;">
        <h2 class="text-center">Login</h2>
        <div class="form-group">
            <label for="loginUsername">Username</label>
            <input type="text" class="form-control" id="loginUsername"
placeholder="Username" required>
        </div>
        <div class="form-group">
            <label for="loginPassword">Password</label>
            <input type="password" class="form-control" id="loginPassword"
placeholder="Password" required>
        </div>
        <button class="btn btn-primary btn-block" onclick="login()">Login</button>
    </div>
</div>

<!-- Bootstrap JS and dependencies (jQuery and Popper.js) -->
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"></scri
pt>
<script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></sc
ript>
</body>
</html>

```

dashboard.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

```



```

<title>Dashboard</title>
<link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">

<script>
  async function logout() {
    try {
      const response = await fetch('php/logout.php', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' }
      });
      const result = await response.json();
      if (result.message === "Logout success!") {
        window.location.href = "index.html";
      } else {
        alert("Error logging out. Please try again.");
      }
    } catch (error) {
      console.error('Logout error:', error);
    }
  }

  async function loadExpenses() {
    try {
      const response = await fetch('php/view_expenses.php');
      const data = await response.json();
      const expenseTableBody = document.getElementById('expenseTableBody');
      expenseTableBody.innerHTML = '';

      if (Array.isArray(data.records) && data.records.length > 0) {
        data.records.forEach(expense => {
          const row = document.createElement('tr');
          row.innerHTML = `
            <td>${expense.created_at}</td>
            <td>${expense.description}</td>
            <td>${expense.amount}</td>
            <td>
              <button class="btn btn-primary btn-sm"
onclick="showEditForm(${expense.id}, '${expense.description}',
${expense.amount})">Edit</button>
              <button class="btn btn-danger btn-sm"
onclick="deleteExpense(${expense.id})">Delete</button>
            </td>
          `;
          expenseTableBody.appendChild(row);
        });
      } else {
        const row = document.createElement('tr');

```

```

        row.innerHTML = '<td colspan="4" class="text-center">No expenses
found.</td>';
        expenseTableBody.appendChild(row);
    }
} catch (error) {
    console.error('Error loading expenses:', error);
}
}

async function addExpense() {
    const description = document.getElementById('description').value;
    const amount = document.getElementById('amount').value;
    try {
        const response = await fetch('php/add_expense.php', {
            method: 'POST',
            body: JSON.stringify({ description, amount }),
            headers: { 'Content-Type': 'application/json' }
        });
        const result = await response.json();
        alert(result.message);
        loadExpenses();
        document.getElementById('addExpenseForm').reset();
    } catch (error) {
        console.error('Error adding expense:', error);
    }
}

function showEditForm(id, description, amount) {
    document.getElementById('expenseId').value = id;
    document.getElementById('editDescription').value = description;
    document.getElementById('editAmount').value = amount;
    document.getElementById('editForm').style.display = 'block';
}

async function updateExpense() {
    const id = document.getElementById('expenseId').value;
    const description = document.getElementById('editDescription').value;
    const amount = document.getElementById('editAmount').value;
    try {
        const response = await fetch('php/edit_expense.php', {
            method: 'POST',
            body: JSON.stringify({ id, description, amount }),
            headers: { 'Content-Type': 'application/json' }
        });
        const result = await response.json();
        alert(result.message);
        document.getElementById('editForm').style.display = 'none';
        loadExpenses();
    }
}

```

```

    } catch (error) {
      console.error('Error updating expense:', error);
    }
  }

  async function deleteExpense(id) {
    try {
      const response = await fetch('php/delete_expense.php', {
        method: 'POST',
        body: JSON.stringify({ id }),
        headers: { 'Content-Type': 'application/json' }
      });
      const result = await response.json();
      alert(result.message);
      loadExpenses();
    } catch (error) {
      console.error('Error deleting expense:', error);
    }
  }
}
</script>
</head>
<body class="bg-light">
  <div class="container mt-5">
    <h1 class="text-center">Welcome to the Dashboard</h1>
    <p class="lead text-center">You are now logged in.</p>
    <div class="text-center mb-4">
      <button class="btn btn-danger" onclick="logout()">Logout</button>
    </div>

    <!-- Form to add a new expense -->
    <div class="mb-4">
      <h4>Add New Expense</h4>
      <form id="addExpenseForm" onsubmit="event.preventDefault(); addExpense();">
        <div class="form-group">
          <label for="description">Description</label>
          <input type="text" id="description" class="form-control"
placeholder="Enter description" required>
        </div>
        <div class="form-group">
          <label for="amount">Amount</label>
          <input type="number" id="amount" class="form-control"
placeholder="Enter amount" required>
        </div>
        <button type="submit" class="btn btn-success">Add Expense</button>
      </form>
    </div>

    <!-- Table to display expenses -->

```

```

<h4>Expense List</h4>
<table class="table table-bordered table-striped">
  <thead class="thead-dark">
    <tr>
      <th>Date</th>
      <th>Description</th>
      <th>Amount</th>
      <th>Actions</th>
    </tr>
  </thead>
  <tbody id="expenseTableBody">
    <!-- Expenses will be loaded here dynamically -->
  </tbody>
</table>

<!-- Form to edit an expense -->
<div id="editForm" style="display: none;">
  <h4>Edit Expense</h4>
  <form onsubmit="event.preventDefault(); updateExpense();">
    <input type="hidden" id="expenseId">
    <div class="form-group">
      <label for="editDescription">Description</label>
      <input type="text" id="editDescription" class="form-control" required>
    </div>
    <div class="form-group">
      <label for="editAmount">Amount</label>
      <input type="number" id="editAmount" class="form-control" required>
    </div>
    <button type="submit" class="btn btn-primary">Update Expense</button>
    <button type="button" class="btn btn-secondary"
onclick="document.getElementById('editForm').style.display='none'">Cancel</button>
  </form>
</div>
</div>

<script src="https://code.jquery.com/jquery-3.5.1.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"></scri
pt>
<script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></sc
ript>

<script>
  document.addEventListener("DOMContentLoaded", loadExpenses);
</script>
</body>

```

</html>

Database\

Database.php

```
<?php
class Database {
    private $host;
    private $username;
    private $password;
    private $database;
    private $connection;

    public function __construct($host, $username, $password, $database) {
        $this->host = $host;
        $this->username = $username;
        $this->password = $password;
        $this->database = $database;
        $this->connect();
    }

    private function connect() {
        $this->connection = mysqli_connect($this->host, $this->username, $this->password, $this->database);

        if (mysqli_connect_errno()) {
            die("Database connection failed: " . mysqli_connect_error());
        }
    }

    public function getConnection() {
        return $this->connection;
    }
}
?>
```

Expense.php

```
<?php
class Expense {
    private $connection;
    private $amount;
    private $description;
    private $id;

    public function __construct($connection) {
        $this->connection = $connection;
```

```

}

public function setAmount($amount) {
    $this->amount = $amount;
}

public function setDescription($description) {
    $this->description = $description;
}

public function setId($id) {
    $this->id = $id;
}

public function getAllExpenses() {
    $query = "SELECT * FROM expenses";
    $result = mysqli_query($this->connection, $query);

    if (!$result) {
        die("Failed to retrieve data: " . mysqli_error($this->connection));
    }

    $expenses = [];
    while ($row = mysqli_fetch_assoc($result)) {
        $expenses[] = $row;
    }
    return $expenses;
}

public function create() {
    $query = "INSERT INTO expenses (amount, description) VALUES ('$this->amount',
'$this->description')";
    $result = mysqli_query($this->connection, $query);

    return $result ? true : false;
}

public function delete() {
    $query = "DELETE FROM expenses WHERE id = '$this->id'";
    $result = mysqli_query($this->connection, $query);

    return $result ? true : false;
}

public function update() {
    $query = "UPDATE expenses SET amount = '$this->amount', description = '$this-
>description' WHERE id = '$this->id'";
    $result = mysqli_query($this->connection, $query);

```

```

        return $result ? true : false;
    }
}
?>

```

User.php

```

<?php
class User {
    public $username;
    public $password;
    public $email;
    public $id;
    private $conn;

    public function __construct($db) {
        $this->conn = $db;
    }

    public function setUsername($username) {
        $this->username = $username;
    }

    public function setPassword($password) {
        $this->password = $password;
    }

    public function setEmail($email) {
        $this->email = $email;
    }

    public function register() {
        $hash_password = password_hash($this->password, PASSWORD_DEFAULT);
        $query = "INSERT INTO users (username, password, email) VALUES (?, ?, ?)";
        $stmt = mysqli_prepare($this->conn, $query);
        mysqli_stmt_bind_param($stmt, "sss", $this->username, $hash_password, $this->email);

        $result = mysqli_stmt_execute($stmt);
        mysqli_stmt_close($stmt);

        return $result;
    }

    public function login() {
        $query = "SELECT id, username, password FROM users WHERE username = ? LIMIT 1";
    }
}

```

```

$stmt = mysqli_prepare($this->conn, $query);
mysqli_stmt_bind_param($stmt, "s", $this->username);
mysqli_stmt_execute($stmt);
$result = mysqli_stmt_get_result($stmt);

if ($row = mysqli_fetch_assoc($result)) {
    if (password_verify($this->password, $row['password'])) {
        $this->id = $row['id'];
        $this->username = $row['username'];
        return true;
    }
}

mysqli_stmt_close($stmt);
return false;
}
}
?>

```

Php\

add\_expense.php

```

<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';

$databse = new Database("localhost", "root", "", "expense_tracker");
$db = $databse->getConnection();
$expense = new Expense($db);

$data = json_decode(file_get_contents("php://input"), true);
$expense->setDescription($data['description']);
$expense->setAmount($data['amount']);

header('Content-Type: application/json');
if ($expense->create()) {
    echo json_encode(["message" => "Expense added."]);
} else {
    echo json_encode(["message" => "Unable to add expense."]);
}
?>

```



#### delete\_expense.php

```
<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';
$databse = new Database("localhost", "root", "", "expense_tracker");
$db = $databse->getConnection();

$data = json_decode(file_get_contents("php://input"), true);
$expense = new Expense($db);
$expense->setId($data['id']);

if ($expense->delete()) {
    echo json_encode(["message" => "Expense deleted."]);
} else {
    echo json_encode(["message" => "Unable to delete expense."]);
}

?>
```

#### edit\_expense.php

```
<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';
$databse = new Database("localhost", "root", "", "expense_tracker");
$db = $databse->getConnection();

$data = json_decode(file_get_contents("php://input"), true);
$expense = new Expense($db);
$expense->setId($data['id']);
$expense->setDescription($data['description']);
$expense->setAmount($data['amount']);

if ($expense->update()) {
    echo json_encode(["message" => "Expense updated."]);
} else {
    echo json_encode(["message" => "Unable to update expense."]);
}

?>
```

#### login.php

```
<?php
session_start();
require_once '../database/Database.php';
require_once '../database/User.php';
```

```

$database = new Database("localhost", "root", "", "expense_tracker");
$db_connection = $database->getConnection();
$user = new User($db_connection);

$data = json_decode(file_get_contents("php://input"));

$user->setUsername($data->username);
$user->setPassword($data->password);

if ($user->login()) {
    $_SESSION['username'] = $user->username; // Corrected here
    echo json_encode(["message" => "Login success"]);
} else {
    echo json_encode(["message" => "Login fail"]);
}
?>

```

logout.php

```

<?php
session_start();
$_SESSION = [];
session_destroy();
header('Content-Type: application/json');
echo json_encode(["message" => "Logout success!"]);
?>

```

register.php

```

<?php
session_start();
require_once '../database/Database.php';
require_once '../database/User.php';

$database = new Database("localhost", "root", "", "expense_tracker");
$db = $database->getConnection();
$user = new User($db);

$data = json_decode(file_get_contents("php://input"));
$user->setUsername($data->username);
$user->setPassword($data->password);
$user->setEmail($data->email);

if ($user->register()) {
    echo json_encode(["message" => "User was registered."]);
}

```

```

} else {
    echo json_encode(["message" => "Unable to register the user."]);
}
?>

```

view\_expenses.php

```

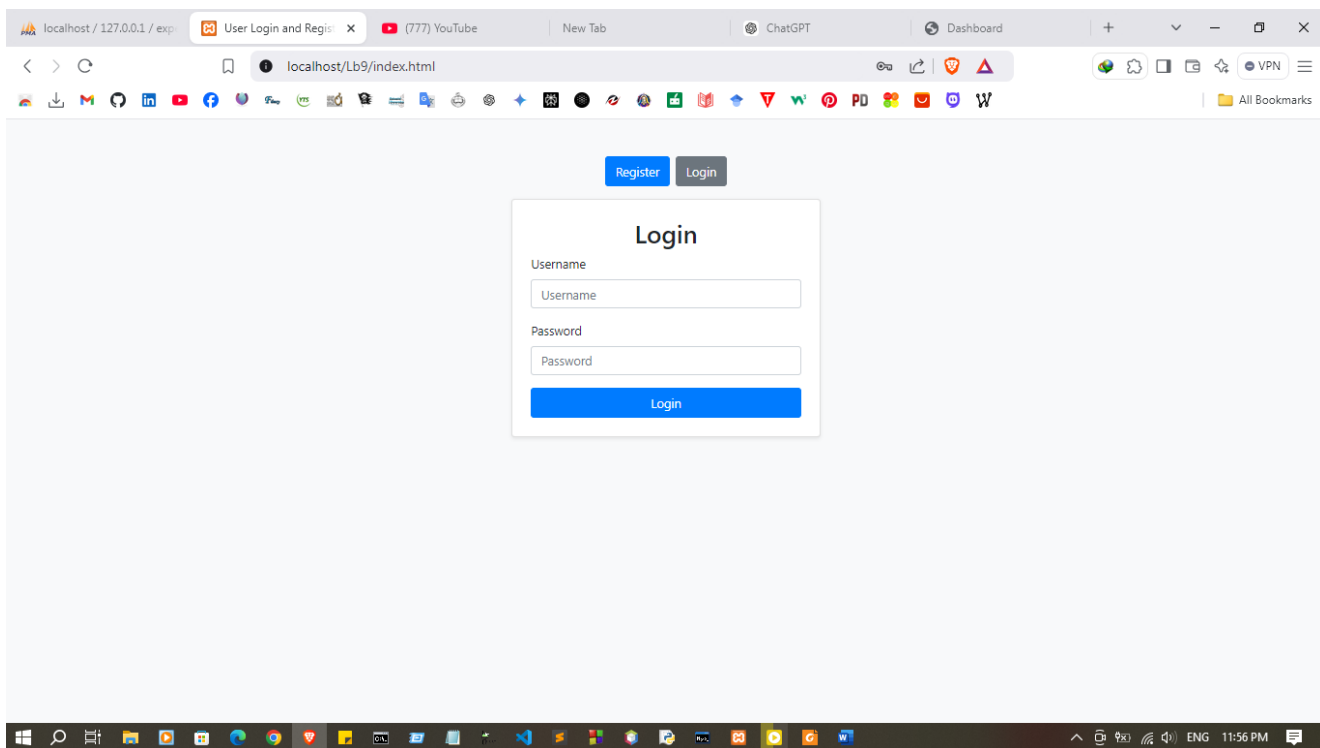
<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';

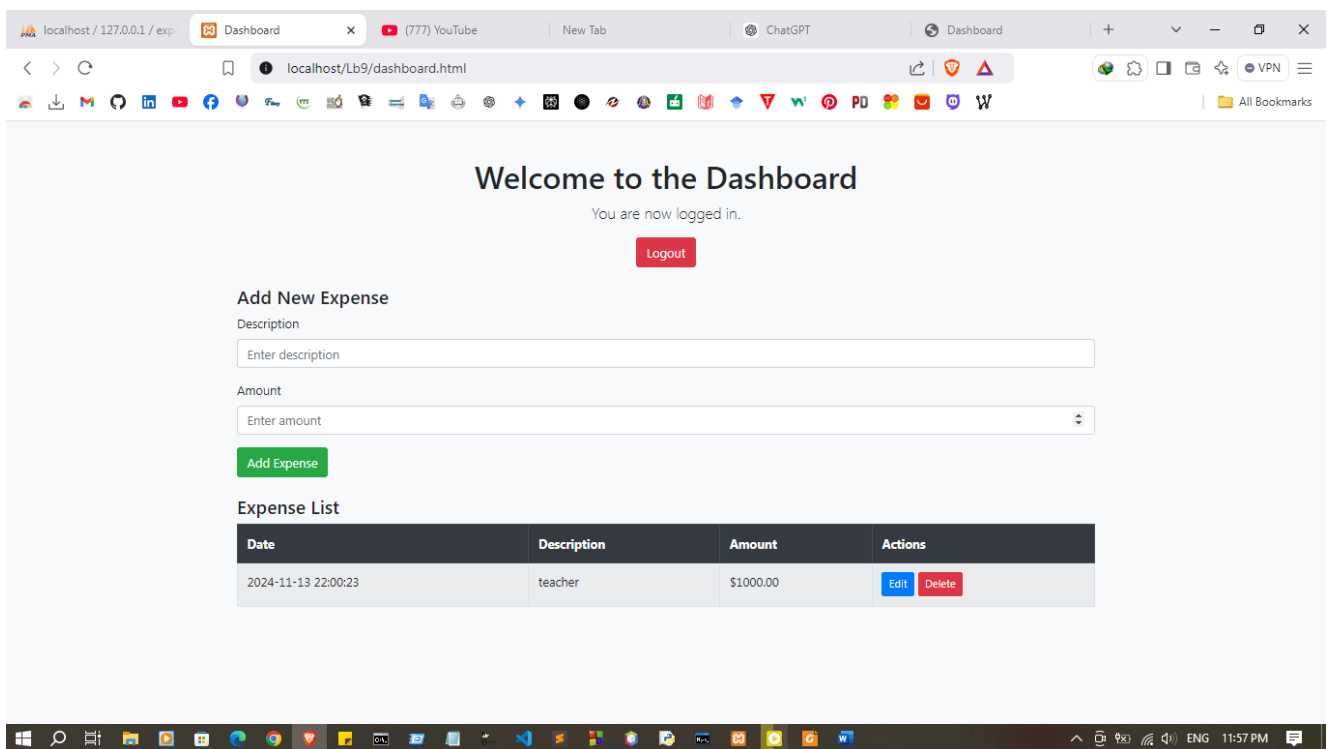
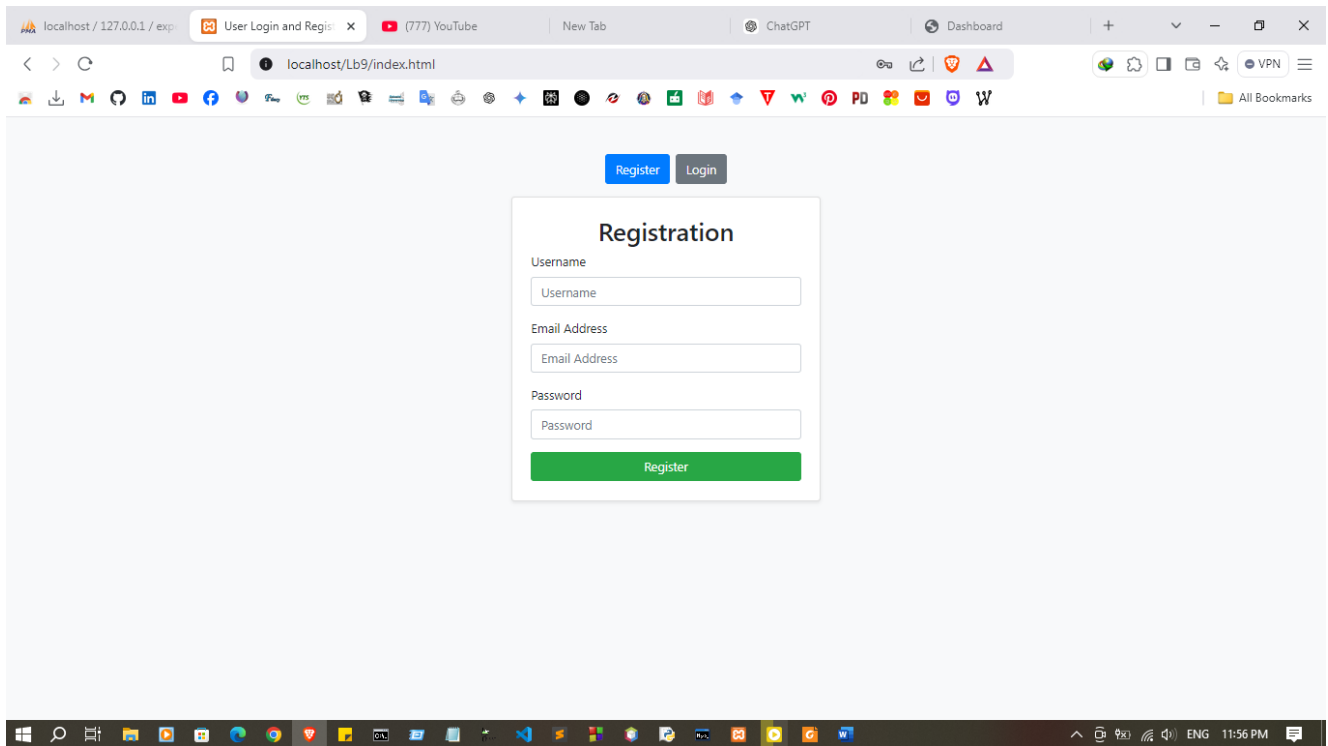
$database = new Database("localhost", "root", "", "expense_tracker");
$db = $database->getConnection();
$expense = new Expense($db);
$expenses = $expense->getAllExpenses();

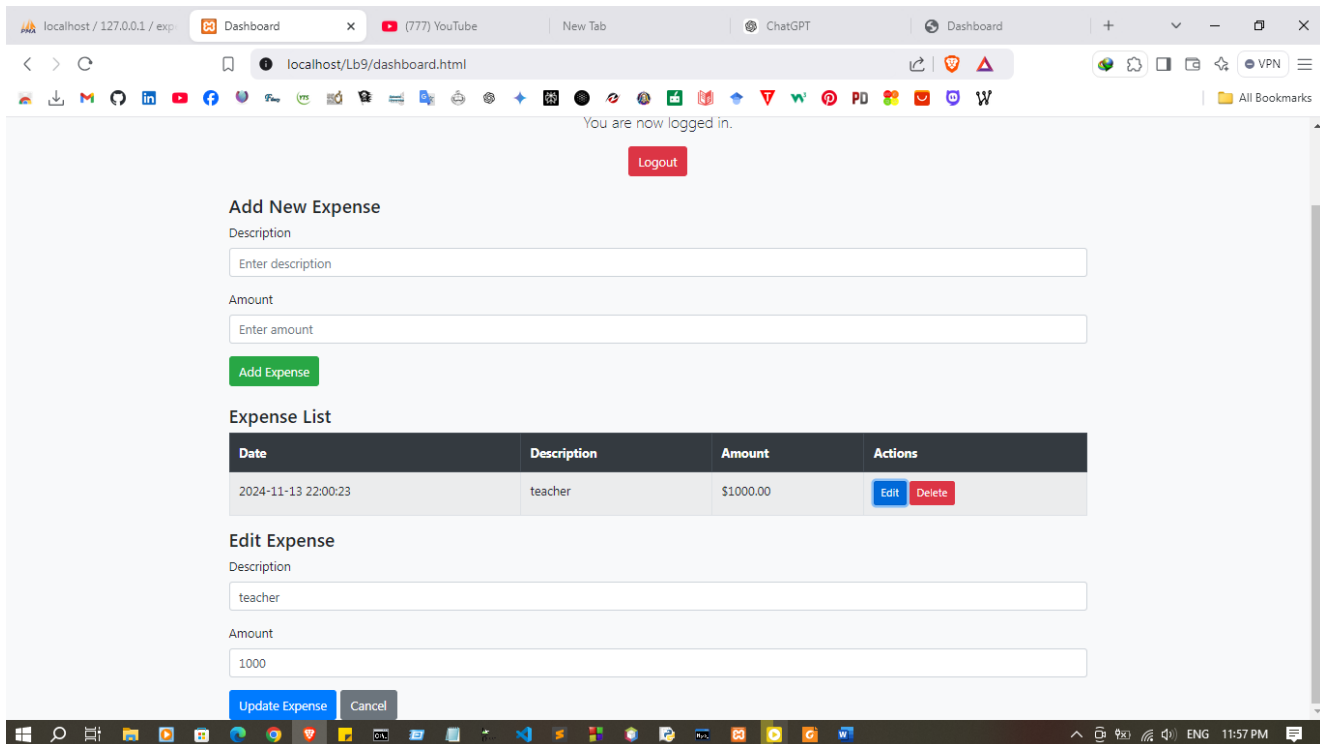
header('Content-Type: application/json');
echo json_encode(["records" => $expenses]);
?>

```

Output







## Discussion

- In this lab session, I worked on implementing CRUD (Create, Read, Update, Delete) operations using PHP and MySQL, with JavaScript on the frontend. The focus was on developing a full cycle of data management functions, where PHP acts as the server-side script to handle requests and interact with the MySQL database, while JavaScript (using the `fetch()` API) manages asynchronous data requests on the frontend. By using `fetch()`, I could send data to PHP scripts without refreshing the page, making the application more dynamic and user-friendly. This lab helped solidify my understanding of PHP as a backend language and how JavaScript can be used to improve user interactions by handling data operations seamlessly in real-time. This experience will be useful in building applications that require user interaction with databases efficiently.